

PLATFORM ENGINEERS • DEPLOYERS

DEVOPS MANUAL

Local dev, build/serve, Cloudflare, VPS, CORS, CI

Open charting PWA — own the axes, swap the engine



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OVERVIEW

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title: "DevOps" description: "Build, serve, Cloudflare, VPS demo, CORS, and CI for the AXIS PWA and Worker."

DEVOPS

Abstract

AXIS DevOps spans five runnable surfaces:

1. **Vite + Solid PWA** — product UI (`bun run dev`)
2. **Static server** (`axis_pwa_server.py` or Vite preview) — production-like assets
3. **Cloudflare Worker** (`worker/`) — edge API / WS / OAuth / on-chain proxy
4. **Desktop (Tauri 2)** — native shell around the same UI (`bun run desktop:dev`)



5. **AXIS CLI** (`packages/cli`) — install, doctor, setup, secrets, deploy, health

See [Desktop \(Tauri\)](#) and [AXIS CLI](#) for operator entry points.

Python evaluation for desk workflows still often uses the **Flask Pro API** (`make run` on `:5002`). The Worker proxies there when `EXTERNAL_BACKEND` is set.

Conceptual model



Topology cheat sheet

Mode	Command sketch	Ports
Dev AXIS	<code>bun run dev</code> / <code>bun run axis dev</code>	Vite (often <code>:3000</code>)
Dev + Flask	+ <code>make -C ../pyne run</code> (checkout sometimes named <code>pynescrypt</code>)	<code>:5002</code>
Static dist	<code>bun run build</code> + <code>python axis_pwa_server.py</code>	<code>:8081</code>
Worker local	<code>bun run axis dev worker</code> / <code>cd worker && bun run dev</code>	<code>:8787</code>
CF Worker prod	<code>axis deploy</code>	HTTPS <code>workers.dev</code> / custom
Operator	<code>axis doctor</code> · <code>axis setup</code> · <code>axis health</code>	—

Page map

Page	Topic
Local dev	Day-to-day loop
AXIS CLI	Install, doctor, setup, secrets, deploy, health
Build and serve	Vite build, SPA server, assets
Cloudflare®	Pages + Worker deploy
Desktop (Tauri)	Native shell
VPS demo	Single-box static + Flask
CORS and origins	Allowed origins matrix
CI and testing	GitHub Actions, coverage, e2e



Frozen names

Surface	Value
Pages project	axis
Pages URL	https://axis.hoox.sh
Worker script	worker-axis
Worker URL	https://worker.axis.hoox.sh
Health service	worker-axis

See also

- [Worker](#)
- [Testing reference](#)
- [AXIS hub](#)

LOCAL DEV

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title: "Local development" description: "Day-to-day AXIS loop: Vite, Flask, Worker wrangler, endpoints, and plugin examples."

LOCAL DEVELOPMENT

Abstract

Local AXIS development is a **multi-process** topology. AXIS is a Solid app under Vite; evaluation usually hits Flask; optional cloud/storage/stream features hit the Worker.



Conceptual model

```
flowchart TB
    subgraph Browser
        UI[Vite HMR app]
    end
    UI -->|engine server| FLASK[Flask :5002]
    UI -->|cloud / run proxy| WRK[wrangler :8787]
    WRK --> FLASK
    UI -->|optional direct| VENUE[Venue REST/WS]
```

Interface surface — commands

Bootstrap (CLI-first)

```
axis install    # app + worker + CLI
axis doctor     # missing worker/wrangler.toml is a warning until `axis setup`
```

Frontend (preferred product path)

```
bun run dev      # Vite + Solid → typically http://127.0.0.1:3000
# or: bun run axis dev
```

Desktop (optional Tauri shell)

```
bun run desktop:dev  # Vite + native window (see devops/desktop)
# or: bun run axis dev desktop
```

Flask Pro API (engine backend)

From the pyne checkout (sister repo; clone [hoox-sh/pyne](#) as `../pyne` — the local directory is sometimes still named `pynescrypt`):

```
make -C ../pyne run  # Flask → :5002
```

Set PWA endpoint to `http://127.0.0.1:5002` for the **server** engine. Workers Manager and Settings can probe this URL.

Worker

```
bun run axis dev worker  # http://127.0.0.1:8787
# or: cd worker && bun run dev
```

Point cloud storage / Worker endpoint at `http://127.0.0.1:8787`. For `/api/run` proxy, set Worker `EXTERNAL_BACKEND=http://127.0.0.1:5002` (works on local wrangler; not on Cloudflare® edge).



AXIS CLI

Preferred operator entry for install / doctor / setup / deploy — see [AXIS CLI](#):

```
bun run axis install
bun run axis doctor
bun run axis setup
bun run axis dev worker
```

Makefile helpers

Target	Intent
python axis_pwa_server.py	Serve built dist/ on :8081 (not Vite)
make worker-dev	Wrangler dev
make worker-install	Install worker deps
bun run test:unit	Bun unit tests under tests/

For the **Solid** product, prefer `bun run dev` / `bun run build` over the legacy `server.ts` unless you are debugging the old tree ([legacy shell](#)).

Internals — useful paths

Path	Role
src/index.tsx	App entry
vite.config.*	Vite / Solid plugin
src/engines/catalog.ts	Default engine endpoint
worker/wrangler.toml	Local vars
public/plugins/	Example modules served at /plugins/

Worked loop

1. Terminal A: `make run` (Flask).
2. Terminal B: `bun run dev`.
3. Open Vite URL; set engine **Server-Side**, endpoint `http://127.0.0.1:5002`.
4. Load `mock-walk` if venues are blocked.
5. Optional Terminal C: Worker with `ALLOW_OPEN_KEYS=1` for cloud library experiments.

Load an example plugin

With Vite or static server:

```
http://127.0.0.1:3000/plugins/example-coingecko-source.js
```

Manager → Plugins → Load from URL.



Invariants & edge cases

- 1. **CORS** — Flask must allow the Vite origin; Worker `pickOrigin` allows `:3000` and `:8081`.
- 2. **Pyodide local** — needs vendor wheels under `public/` (copied to dist on build).
- 3. **Two endpoints** — topbar endpoint for engine vs cloud storage plugin config can differ (Flask vs Worker).
- 4. `make` prints the Vite URL `:3000`. Compose serves the built PWA on `:8081`.

Failure modes

Issue	Fix
Engine not ready	Flask down / wrong port
CORS errors	Align origins; see CORS
Plugin 404	File only under <code>src/plugins/</code> not <code>public/plugins/</code>
Worker scripts 401	Set open keys or mint <code>pn_</code> key

See also

- [Build and serve](#)
- [Worker local](#)
- [Plugin examples](#)

CLI

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title: "AXIS CLI" description: "packages/cli — install, doctor, Worker setup (D1/OAuth), secrets, deploy, health — CLI-first operator path."

AXIS CLI

Abstract

The **AXIS CLI** (`@hoox-sh/axis-cli` under `packages/cli`) is the **CLI-first** operator entry for local install, Worker bootstrap, Cloudflare® secrets, deploy, and health checks. It wraps Bun + Wrangler with AXIS-specific paths and defaults (Worker name `worker-axis`, D1 `pynescrypt`, OAuth device flow).

 `axis --help` listing install, doctor, setup, deploy, health



Requires **Node ≥ 20** to run the published binary (`npm i -g @hoox-sh/axis-cli`) and **Bun ≥ 1.2** for install/dev/build commands that shell out to Bun. Product version follows root `package.json` ; the CLI package (`@hoox-sh/axis-cli`) is versioned independently and publishes on `v*` tags via `.github/workflows/release.yml` .

Conceptual model

```
flowchart LR
    CLI[axis CLI]
    CLI --> APP[app deps / Vite]
    CLI --> WRK[worker/ wrangler]
    CLI --> CF[Cloudflare edge]
    WRK --> D1[(D1 pyonescript)]
    WRK --> SECRETS[secrets / vars]
```

Install

```
# npm
npm install -g @hoox-sh/axis-cli
axis --help

# from monorepo root
bun install
cd packages/cli && bun install && bun run build && cd ../../

bun run axis --help
# Make pass-through
make axis ARGS="--help"
# Optional global link
cd packages/cli && bun link && axis --help
```

Repo aliases (bun scripts & Make targets)

`axis` is the primary interface. These aliases exist for working inside the repo checkout without a global install:

Alias	Runs
<code>bun run axis -- <args></code>	<code>packages/cli/bin/axis.js</code>
<code>bun run axis:install</code>	<code>axis install</code>
<code>bun run axis:doctor</code>	<code>axis doctor</code>
<code>bun run axis:setup</code>	<code>axis setup</code>
<code>bun run axis:deploy</code>	<code>axis deploy</code> (default worker)
<code>bun run axis:health</code>	<code>axis health</code>
<code>make axis ARGS="doctor --remote"</code>	same bin with args
<code>make axis-install · axis-doctor · axis-setup · axis-deploy · axis-health</code>	fixed wrappers

MCP (agents): `axis mcp --key pn_...` (stdio) · `axis mcp config` — see [MCP server](#).

When using `bun run axis setup -- --flag` , keep the `--` so Bun forwards flags to the CLI.



Command surface

Command	Role
<code>axis install</code>	<code>bun install</code> root + <code>worker/</code> + CLI
<code>axis doctor [--remote]</code>	Toolchain, CF auth, optional live <code>/health</code> . Missing <code>worker/wrangler.toml</code> is a warning until <code>axis setup</code> (expected on a fresh clone).
<code>axis setup</code>	Bootstrap: install → ensure toml → local D1
<code>axis setup --prod</code>	Production bootstrap: remote D1 schema + <code>API_KEYS</code> KV
<code>axis setup worker</code>	Copy <code>wrangler.toml.example</code> → <code>wrangler.toml</code> if missing
<code>axis setup d1 --local --remote [--create]</code>	Apply <code>worker/schemas/scripts.sql</code> (scripts + version history)
<code>axis setup kv [--usage]</code>	Create and bind <code>API_KEYS</code> KV (required when D1 is on)
<code>axis setup oauth --github-client-id ...</code>	Set public OAuth App id in <code>[vars]</code> (or <code>--secret</code>)
<code>axis keys create [--tier hobby]</code>	Mint a <code>pn_...</code> key via <code>POST /api/keys</code> (<code>ADMIN_TOKEN</code>)
<code>axis keys validate --key pn_...</code>	Validate a key against the Worker
<code>axis secret put list delete</code>	Wrangler secrets (<code>ADMIN_TOKEN</code> , <code>EXTERNAL_BACKEND</code> , ...)
<code>axis deploy / deploy worker</code>	Apply remote D1 schema, then deploy Worker <code>worker-axis</code>
<code>axis deploy pages</code>	Vite build + Pages project
<code>axis deploy all</code>	Schema + Worker + Pages
<code>axis health [--oauth] [--scripts] [--url ...]</code>	Probe <code>/health</code> , optional OAuth start, optional <code>/api/scripts</code>
<code>axis mcp [--key pn_...]</code>	Stdio MCP proxy to Worker <code>POST /mcp</code>
<code>axis mcp config [--stdio]</code>	Print MCP client JSON (remote URL or stdio spawn)
<code>axis whoami</code>	Cloudflare® account
<code>axis dev</code>	Vite product UI
<code>axis dev worker</code>	Local wrangler Worker
<code>axis dev desktop</code>	Tauri desktop shell

Global flags: `--json`, `--quiet`, `-y` / `--yes`.

Installation self-check

Every `axis` command starts with a lightweight, local-only self-check (skipped for `--version` / `--help`, silenced by `--quiet` / `--json`). It never blocks a command — it only prints stderr warnings when something is off:

- **Runtime compliance** vs `engines` (Node ≥ 20, Bun ≥ 1.2)
- **Install context** — npm install, npx cache, or repo checkout
- **Version drift** — a globally installed CLI older than the repo checkout it is operating on (`npm i -g @hoox-sh/axis-cli` to update)



The same check appears as the optional **CLI installation** row in `axis doctor` (and in `doctor --json` as `cli-install`).

 `axis doctor`: required toolchain and wrangler checks passed

Production checklist

```
axis install
axis doctor
axis setup --prod --github-client-id 0v23li...
axis secret put ADMIN_TOKEN # comments empty [vars] ADMIN_TOKEN, deploys, then sets secret
axis secret put EXTERNAL_BACKEND
axis deploy all
axis keys create # --admin-token / AXIS_ADMIN_TOKEN / prompt
axis health --scripts --oauth
axis mcp config --key pn_... # MCP client snippet (see [MCP server](/axis/docs/worker/mcp))
# Paste Worker URL + pn_... into Settings -> Script storage
```

Item	Guidance
GITHUB_OAUTH_CLIENT_ID	Public OAuth App id; Device Flow enabled on GitHub; env wins over body <code>clientId</code>
GITLAB_OAUTH_CLIENT_ID	Same for GitLab when using git storage
ADMIN_TOKEN / EXTERNAL_BACKEND	Prefer <code>axis secret put</code> . An empty <code>[vars]</code> stub collides with secrets (Cloudflare 10053)
API_KEYS KV	<code>axis setup kv</code> then <code>axis deploy worker</code> . D1 without KV fails closed (<code>API_KEYS_REQUIRED</code>)
ALLOW_OPEN_KEYS	"0" in production
REQUIRE_RUN_AUTH	"1" to force Bearer on <code>/api/run</code> even without KV
Worker script	<code>worker-axis</code> (https://worker.axis.hoox.sh)
Pages project	<code>axis</code> (https://axis.hoox.sh)

Env overrides

Variable	Role
AXIS_ROOT	Force monorepo root
AXIS_WORKER_URL	Default health / deploy probe URL
AXIS_ADMIN_TOKEN	Admin token for <code>axis keys create</code>
AXIS_API_KEY	Default key for <code>axis keys validate</code> / <code>axis mcp</code>
CLOUDFLARE_API_TOKEN	Non-interactive Wrangler auth
AXIS_CLI_SRC=1	Load <code>src/</code> instead of <code>dist/</code>



Internals

Path	Role
packages/cli/bin/axis.js	Bin entry (Bun)
packages/cli/src/commands/*	Commander handlers (install , doctor , setup , keys , deploy , secrets , health , dev , whoami , mcp)
packages/cli/src/services/wrangler-toml.ts	Toml vars, D1 id, KV binding upsert
packages/cli/src/services/health.ts	/health + OAuth probes
Root package.json axis / axis:* scripts	Convenience wrappers
Root Makefile axis / axis-*	Make wrappers

See also: [Installation](#), [MCP \(agents\)](#), [MCP server](#), [Cloudflare® deployment](#), [Worker bindings](#), [Worker auth](#), [packages/cli/README.md](#).

BUILD AND SERVE

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title: "Build and serve" description: "Vite production build, dist layout, axis_pwa_server.py SPA rules, and asset caching."

BUILD AND SERVE

Abstract

Production AXIS assets are a **static site** produced by Vite (`bun run build` → `dist`). Serve them with any static host, Cloudflare Pages, or the repo's `axis_pwa_server.py` SPA-aware server.

Critical requirement: **real binary assets** for Pyodide (`.wasm` , `.whl` , `.zip`) must not be rewritten to `index.html` .



Conceptual model

```
flowchart LR
    SRC[src/** Solid]
    PUB[public/**]
    SRC --> VITE[vite build]
    PUB --> VITE
    VITE --> DIST[dist/]
    DIST --> PAGES[CF Pages]
    DIST --> PWA[axis_pwa_server.py]
    DIST --> PREV[vite preview]
```

Interface surface

package.json scripts

```
"dev": "vite",
"build": "vite build",
"preview": "vite preview",
"test": "bun test tests/ worker/tests/",
"test:coverage:gate": "bun run test:coverage && bun scripts/check-coverage.mjs 95",
"test:security": "bun test tests/security/",
"test:e2e:smoke": "playwright test --grep @smoke",
"test:all": "bun run test:coverage:gate && bun run test:security"
```

axis_pwa_server.py

```
bun run build
python axis_pwa_server.py
# HOST / PORT env; default 0.0.0.0:8081 → dist/
```

Behavior (axis_pwa_server.py):

Rule	Detail
Root	Serves dist/
Cache	/assets/* → immutable long cache; else no-cache
GET /health (/healthz)	200 text/plain ok — never SPA-fallback to index.html
OPTIONS	204 with CORS headers (not 501)
/favicon.ico	Serves assets/icon-192.png when present
SPA fallback	Unknown paths → /index.html
Never rewrite	/health , /favicon.ico , /assets/ , /plugins/ , /vendor/ , /pyodide/ , or extensions including .js/.css/.whl/.wasm/.zip/.py/.json...

This SPA exception list exists because Pyodide + micropip die with opaque `BadZipFile` when HTML is returned for a wheel URL. The engine's `assertZipAsset` detects that class of failure early.



Expected public assets

Path (public → dist)	Purpose
/plugins/*.js	Example dynamic plugins
/vendor/*.whl	pynescrypt + antlr wheels
/pyodide/v0.26.2/**	Self-hosted Pyodide
/pyodide/pynescrypt_runtime.py	Browser run bridge

Internals

Path	Role
package.json	Build scripts
axis_pwa_server.py	Threading HTTP SPA server
src/engines/catalog.ts	Asset URL expectations
LEGACY.md	Product path vs old shell

Makefile note

`make pages-deploy` may historically deploy the wrong tree — **prefer:**

```
bun run build
wrangler pages deploy dist --project-name=axis
```

Worked example — static desk demo

```
bun install
bun run build
PORT=8081 python axis_pwa_server.py
# another terminal
make run # Flask :5002
```

Open `http://127.0.0.1:8081`, engine endpoint `http://127.0.0.1:5002`.

Invariants & edge cases

1. **Service workers / offline** — product path uses Vite PWA assets; legacy root `sw.js` is not the shipping SW.
2. **Relative pyodide index** — resolves against `location.origin`.
3. **Do not hand-edit dist** — regenerate from build.



Failure modes

Symptom	Cause
Blank routes on refresh	Server lacks SPA fallback
Pyodide HTML wheel error	Fallback rewriting <code>.whl</code> or missing vendor
Missing plugins	<code>public/plugins</code> not copied (check Vite publicDir)

See also

- [Local dev](#)
- [Cloudflare](#)
- [Engines](#)

DESKTOP

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title: "Desktop (Tauri)" description: "Run and package AXIS as a native desktop app with Tauri 2."

DESKTOP (TAURI)

Abstract

AXIS ships as a browser PWA and as an optional **desktop shell** built with [Tauri 2](#). The Solid/Vite UI is unchanged; Tauri embeds it in a system webview (WebKitGTK on Linux, WKWebView on macOS, WebView2 on Windows).

Commands

From the repo root (after `bun install`):

```
bun run desktop:dev      # Vite :3000 + Tauri window (hot reload)
bun run desktop:build    # Vite production build + native installers
bun run desktop:info     # toolchain / webview diagnostics
```

Script	What it does
<code>desktop:dev</code>	Runs <code>tauri dev</code> → <code>beforeDevCommand</code> starts Vite, opens a native window on <code>http://127.0.0.1:3000</code>
<code>desktop:build</code>	Runs <code>tauri build</code> → <code>bun run build</code> then packages <code>dist/</code>
<code>desktop:info</code>	Prints Rust, system libraries, and webview status

Installers land under `src-tauri/target/release/bundle/` (AppImage, deb, rpm, msi, dmg — depending on host OS and `bundle.targets`).

CI (GitHub Actions)

Every **push to main** (and PRs that touch desktop/frontend paths) runs `.github/workflows/desktop.yml` :

Trigger	Result
Push <code>main</code>	Matrix build: Linux, macOS arm64, macOS x64, Windows → workflow artifacts (14-day retention)
Tag <code>v*</code> / <code>desktop-v*</code>	Same matrix + GitHub Release assets attached to the tag
<code>workflow_dispatch</code>	Manual re-run



Artifacts are named `axis-desktop-{linux|macos-arm64|macos-x64|windows}-{sha}`. Concurrency cancels superseded runs on the same ref so each push rebuilds cleanly.

Prerequisites

Always

- [Bun](#) (or another JS package manager that can run the scripts)
- [Rust](#) (`rustc` / `cargo`) — see `src-tauri/Cargo.toml` `rust-version`
- System webview libraries for your platform ([Tauri prerequisites](#))

Linux (Arch / CachyOS example)

```
sudo pacman -S --needed webkit2gtk-4.1 base-devel curl wget file \
  openssl appmenu-gtk-module libappindicator-gtk3 librsvg patchelf
```

Debian/Ubuntu equivalents use `libwebkit2gtk-4.1-dev` and related `-dev` packages (see the Tauri docs for the current list).

Engine backends

Desktop AXIS is the same product as the PWA. For Pine evaluation you still need one of:

Engine	Notes
Local pyne Pro API	Clone hoox-sh/pyne as <code>../pyne</code> (<code>make run</code> → <code>:5002</code> ; Vite proxies <code>/run</code> in dev). Local checkouts are sometimes still named <code>pynescrypt</code> .
Cloudflare Worker	Configure worker URL in Manager
Pyodide (in-webview)	Fully offline path; largest first load

Layout

Path	Role
<code>src-tauri/</code>	Rust host, <code>tauri.conf.json</code> , icons, capabilities
<code>src-tauri/tauri.conf.json</code>	App id <code>sh.hoox.axis</code> , window size, build hooks
<code>src/</code>	Unchanged Solid product UI
<code>dist/</code>	Vite output consumed by <code>frontendDist</code>

Behaviour notes

- **Service worker** is skipped in the Tauri shell (`src/pwa/register-sw.ts`). Offline install is a PWA concern; the desktop app is already installed as a native binary.
- **Window** defaults: 1440×900, min 960×640, centered.
- **CSP** is left open (`null`) so external venues, Pyodide assets, and worker APIs work the same as in the browser. Tighten later if you ship a locked-down build.
- Icons are generated from `public/assets/icon-512.png` via `bunx tauri icon`.



Native menu & open from disk

The Tauri host builds a native app menu:

Menu	Item	Action
File	Open Script... (⌘/Ctrl+O)	System multi-file dialog → library + editor tabs
File	Quit AXIS	Quit (platform predefined)
Help	About AXIS	Info dialog (version / license / site)

Flow

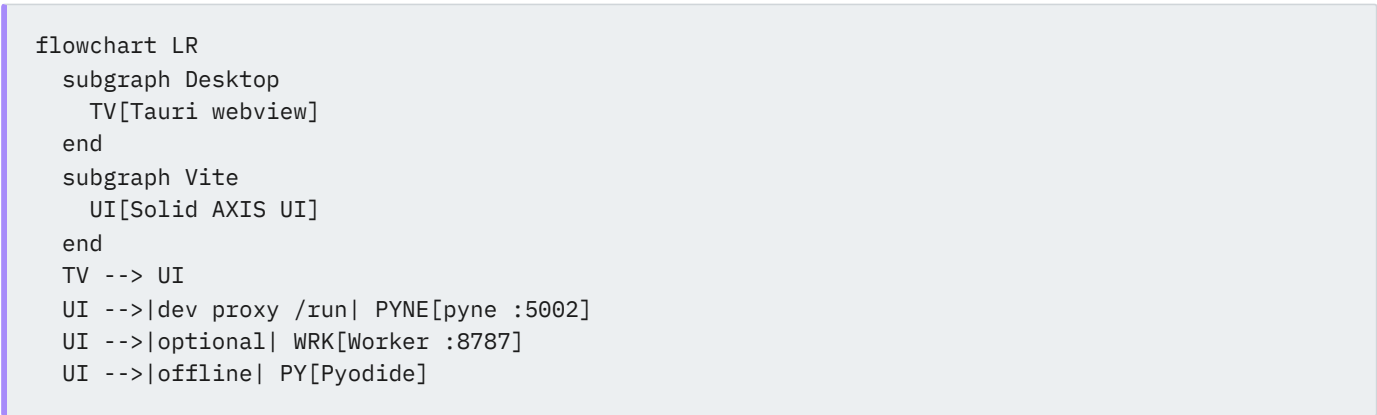


Path	Role
src-tauri/src/lib.rs	Menu, open_pine_scripts command, dialog + disk read
src/desktop/	isTauriShell , menu listen, open + About wiring
src/storage/import-pyne-open.ts	Shared status/logs/editor open after import
src/storage/import-pyne-files.ts	importPyneSources (text) + importPyneFiles (browser File)

Accepted extensions match the PWA drop path: .pyne , .pine , .pinescript , .pinev5 , .pinev6 . Per-file size cap on the host: **8 MiB**.

Drag-and-drop of the same extensions still works in the desktop webview.

Conceptual model



CLOUDFLARE

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title: "Cloudflare deployment" description: "Pages + Worker for AXIS: Pages axis.hoox.sh (project axis), Worker worker.axis.hoox.sh, CLI deploy, bindings, security checklist."

CLOUDFLARE® DEPLOYMENT

Abstract

Production AXIS on Cloudflare® is a **split deploy**:



Piece	Role
Pages	Static PWA (<code>dist/</code>)
Worker	JSON API + WebSocket (<code>worker/</code>)

Pages: project `axis` , canonical `https://axis.hoox.sh` . **Worker:** script `worker-axis` at `https://worker.axis.hoox.sh` .

Evaluation truth: Worker `/api/run` proxies to **EXTERNAL_BACKEND** unless the gated Pyodide path is fully implemented and enabled. You still need a Python host (VPS Flask, container, etc.) for real Pine Script™ fidelity today.

Prefer the **AXIS CLI** for setup and deploy (`axis deploy`).

Conceptual model

```
flowchart TB
    USER[Browser]
    USER --> PAGES[CF Pages dist]
    USER --> WRK[CF Worker]
    WRK -->|EXTERNAL_BACKEND| VPS[Flask / PYNE host]
    WRK --> KV[KV]
    WRK --> D1[D1]
    WRK --> D0[SessionD0]
```

Deploy procedure

1. Provision bindings (once)

See [Worker bindings](#):

- KV `API_KEYS` , `USAGE`
- D1 `pynscript` + schema
- Optional R2, Durable Objects

2. Configure secrets / vars

Production checklist:



Setting	Production guidance
EXTERNAL_BACKEND	Public HTTPS URL of Flask/PYNE API (<code>axis secret put</code>)
ALLOWED_ORIGIN	Extra exact origins (comma-separated); product hosts + <code>*.axis.pages.dev</code> are built-in — CORS
ADMIN_TOKEN	Secret, strong
API_KEYS KV	Bind in prod — D1 without KV fails closed (<code>API_KEYS_REQUIRED</code>)
ALLOW_OPEN_KEYS	<code>"0"</code> or unset
REQUIRE_RUN_AUTH	<code>"1"</code> optional force Bearer on <code>/api/run</code>
GITHUB_OAUTH_CLIENT_ID	Env preferred; never rely on body <code>clientId</code> alone
PYODIDE_IN_WORKER	<code>"disabled"</code> until wheel pipeline ready

3. Deploy Worker

```
# Preferred: AXIS CLI
axis deploy
# or: make axis-deploy

# Schema first (remote)
axis setup d1 --remote

# Equivalent raw wrangler
cd worker
bun install
bun run deploy
```

4. Build & deploy Pages

```
bun run axis deploy pages
# or
bun run build
bunx wrangler pages deploy dist --project-name=axis
# Make: make pages-deploy
```

5. Point the PWA

- Engine endpoint: Worker URL (if path-mapped to `/run`) **or** Flask URL directly
- Cloud storage endpoint: Worker origin
- Ensure CORS allows Pages origin



Internals

Path	Role
<code>worker/wrangler.toml</code> / <code>.example</code>	Worker name + bindings
<code>worker/README.md</code>	Ops narrative
<code>worker/package.json</code>	deploy scripts
<code>packages/cli/</code>	<code>axis deploy</code> / <code>axis health</code>
Makefile <code>axis-deploy</code> / <code>pages-deploy</code> / <code>worker-deploy</code>	Convenience targets

Health check

```
axis health
# or
curl -sS https://<worker>/health
# service: worker-axis
```

Invariants & edge cases

1. **Pages ≠ Worker host** — CORS required unless same-origin routing via custom domain routes.
2. **Localhost EXTERNAL_BACKEND** is useless from the edge.
3. **DO migrations** apply on deploy when uncommented.
4. **Observability** enabled in wrangler for log tails (`wrangler tail`).

Failure modes

Issue	Fix
503 NO_BACKEND	Set reachable EXTERNAL_BACKEND
CORS blocked	ALLOWED_ORIGIN mismatch
SPA HTML for wheels	Pages must serve <code>/vendor</code> and <code>/pyodide</code> as static files
Stream NO_DO	Enable SessionDO bindings

See also

- [Worker](#)
- [CORS](#)
- [VPS demo](#) (backend co-host)

VPS DEMO

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title: "VPS demo topology" description: "Single-box demo: static AXIS dist + Flask Pro API (+ optional reverse proxy) without Cloudflare."

VPS DEMO TOPOLOGY

Abstract

A **VPS demo** is the lowest-surprise production-like setup for Pine evaluation: one machine runs Flask (PYNE) and serves the static AXIS build. Optional nginx/caddy terminates TLS and reverse-proxies `/run` same-origin to avoid CORS.

No Worker required. Cloud storage and DO streams will not work without the Worker.

Conceptual model

```

flowchart LR
    U[Users]
    U --> NGX[nginx/caddy :443]
    NGX -->|/| STATIC[dist]
    NGX -->|/run| FLASK[Flask :5002]
  
```

Interface surface

Processes

Process	Command	Port
Build (CI or once)	<code>bun run build</code>	—
Static	<code>PORT=8081 python axis_pwa_server.py</code>	8081
API	<code>make run / python -m backend.app</code>	5002

Same-origin proxy sketch (Caddy)

```

axis.example.com {
  handle /run* {
    reverse_proxy 127.0.0.1:5002
  }
  handle /optimize* {
    reverse_proxy 127.0.0.1:5002
  }
  handle {
    reverse_proxy 127.0.0.1:8081
  }
}
  
```

Then PWA endpoint = `https://axis.example.com` and the **server** engine posts to `/run` on the same host.

Production (split: Pages PWA + Hetzner API)

- **PWA** — Cloudflare Pages custom domain `https://axis.hoox.sh`
- **PYNE Pro API** — Hetzner VPS `https://pynescrypt.online` (Cloudflare orange-cloud → origin `204.168.138.51`). `nginx :443` reverse-proxies `/run`, `/optimize`, `/ws/`, `/health`, `/lsp/`, `/datafeed/`, ... to `gunicorn 127.0.0.1:5002`
- PWA **Backend URL** = `https://pynescrypt.online` (cross-origin; product CORS regex)
- Cloudflare Pages previews (`*.axis.pages.dev`) use the same Backend URL — pyne must allow those Origins on `/health` and `/run` (always-on product regex; see [CORS](#))

SSH alias: `ssh pynescrypt` (`204.168.138.51`). The previous combined VPS (`ssh axis / 162.254.38.194`) stays until decommission.

Update deploy (from a machine that can SSH to the VPS):



```
# 1. Ship latest main
git push origin main

# 2. On VPS (paths may vary – often /root/axis + rsync dist → PWA WorkingDirectory)
# Pro API (from pynescrypt checkout)
./scripts/deploy_vps.sh
# or:
ssh pynescrypt 'cd /root/pynescrypt && git pull --ff-only && systemctl restart pynescrypt-api'
# PWA: `bun run build` + wrangler pages deploy (axis.hoox.sh)
```

Local operator CLI for the **Worker** edge (separate from VPS static): `axis deploy` — see [AXIS CLI](#).

Hardened network (UFW) + health checks

Production VPS pattern (verified on `pynescrypt.online`):

Surface	Bind / allow	Health URL
UFW	deny incoming default; allow 22/tcp, 80/tcp, 443/tcp	—
nginx	0.0.0.0:443 TLS (+ :80 ACME)	<code>https://pynescrypt.online/health</code> → proxy <code>127.0.0.1:5002/health</code>
Pro API (gunicorn)	<code>127.0.0.1:5002</code> only	loopback; not public
PWA static	Cloudflare Pages (<code>axis.hoox.sh</code>)	not on this VPS
fail2ban	sshd jail	—

Do not open UFW for `5002`, `80`, or `8787` on a public VPS. Direct probes to `http://<vps-ip>:5002/health` **must time out** if hardening is correct.

Workers Manager browser probes:

- On **product API host** (`pynescrypt.online`), pyne Pro is probed via **same-origin** `/health` (nginx), not `http://127.0.0.1:5002` (that would hit the *client* PC and always look “down”).
- From **CF Pages** (`axis.hoox.sh` / `*.axis.pages.dev`) catalog `publicEndpoint` (`https://pynescrypt.online`) is used — never client loopback.
- CF Worker health stays on `https://worker.axis.hoox.sh/health` (edge; independent of VPS UFW).

Minimal without TLS

- Static `:8081`, Flask `:5002`
- Prefer a hostname over raw IP when the page is HTTPS
- Open firewall; accept CORS from static origin (configure Flask accordingly)



Internals

Concern	Notes
Pyodide offline	Ship full <code>dist</code> including <code>vendor</code> + <code>pyodide</code>
Process supervisor	systemd units for flask + <code>axis_pwa_server</code>
Updates	Rebuild dist artifact; restart static only
Secrets	Flask admin / pro keys separate from Worker

Worked example — systemd sketch

```
# /etc/systemd/system/axis-pwa.service
[Service]
WorkingDirectory=/opt/axis
Environment=PORT=8081
ExecStart=/usr/bin/python3 axis_pwa_server.py
Restart=on-failure
```

Pair with an existing `backend` unit for Flask.

Invariants & edge cases

1. **CPU/RAM** — Pyodide is browser-side; VPS load is Flask evaluate concurrency.
2. **Do not expose open admin tokens.**
3. **Disk** — dist + wheels are tens of MB.
4. Hybrid: VPS Flask as `EXTERNAL_BACKEND` for a CF Worker still works if VPS has a public URL.

Failure modes

Issue	Fix
Mixed content	HTTPS page calling HTTP Flask blocked
CORS	Prefer same-origin proxy
502 on /run	Flask crashed; check journalctl

See also

- [Build and serve](#)
- [Cloudflare](#)
- [CORS](#)

CORS AND ORIGINS

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title: "CORS and origins" description: "Worker pickOrigin: local-dev, product hosts, project-scoped Pages previews, ALLOWED_ORIGIN list, Flask constraints."

CORS AND ORIGINS

Abstract

AXIS is a browser product that talks to **cross-origin** backends (Flask, Worker, venues). CORS misconfiguration is the #1 local-deploy footgun after missing wheels.

The Worker implements an explicit origin picker (`pickOrigin` in `worker/src/index.ts`); Flask and venue APIs have their own rules.

Conceptual model

```

flowchart TD
    ORIGIN[Browser Origin header]
    ORIGIN --> W[Worker pickOrigin]
    W -->|localhost / 127.0.0.1| ECH01[Echo]
    W -->|product hosts| ECH02[Echo]
    W -->|*.axis.pages.dev| ECH03[Echo]
    W -->|ALLOWED_ORIGIN list| ECH04[Echo if listed]
    W -->|else| FALLBACK[First ALLOWED_ORIGIN]
    ORIGIN --> F[Flask CORS middleware]
    ORIGIN --> V[Venue APIs]

```

Worker behavior

Resolution order for `Access-Control-Allow-Origin`:

1. Local-dev (always echoed)

Any `http://` or `https://` origin on `localhost` or `127.0.0.1`, optional port (e.g. Vite `:3000`, `axis_pwa_server` `:8081`, Playwright ephemeral ports).

Not allowed (and not needed): `http://0.0.0.0:...` — browsers do not use `0.0.0.0` as a page origin; binding with `HOST=0.0.0.0` only means “listen on all interfaces.”

Localhost/127 allowlisting is **safe for CORS**: only pages actually served from those hosts present that `Origin`. A public site cannot forge `Origin: http://localhost:3000` in a real browser.

2. Product hosts (2.0.1+)

Exact product / HOOX / legacy PYNE hosts under:

- `*.hoox.sh` / `hoox.sh`
- `*.pynescrypt.online` / `pynescrypt.online`

Examples: `https://axis.hoox.sh` (canonical Pages), `https://hoox.sh`, `https://pynescrypt.online`.

3. Project-scoped Cloudflare® Pages (not open `*.pages.dev`)

Canonical Pages is `https://axis.hoox.sh` (product host above). Preview hashes on `*.axis.pages.dev` (and the apex `axis.pages.dev`) are also echoed. Legacy `*.pynescrypt-axis.pages.dev` still matches. Arbitrary third-party Pages projects (`evil.pages.dev`, other apps) fall through to the allowlist fallback — they are **not** reflected.

4. `ALLOWED_ORIGIN` allowlist

Comma-separated exact origins in `env.ALLOWED_ORIGIN` (e.g. `https://a.example,https://b.example`). If the request Origin is listed, echo it; otherwise return the **first** entry (default `https://pynescrypt.online`).

CORS headers applied:

```

Access-Control-Allow-Origin: <picked>
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type, Authorization, X-Admin-Token, If-Match
Access-Control-Max-Age: 86400
Vary: Origin

```



OPTIONS → 204 preflight.

Production implication

For a custom domain already under product regex, no extra var is required. For one-off preview hosts outside the project, list them:

```
ALLOWED_ORIGIN = "https://my-preview.example.com,https://axis.example.com"
```

Flask / server engine

The **server** engine posts from the browser to the configured endpoint. Flask Pro API must:

1. Answer OPTIONS preflight if cross-origin.
2. Reflect or allow the PWA origin.
3. Accept Content-Type: application/json .

Pyne always appends the same product Origin regex as Worker `pickOrigin` (including `*.axis.pages.dev`) even when systemd `ALLOWED_ORIGINS` is a short list. `GET /health` (AXIS Settings probe) and `POST /run` are **free CORS paths** — they echo any `Origin` so a Pages preview can reach `https://axis.hoox.sh` without a per-preview allowlist entry.

Same-origin reverse proxy (VPS demo) eliminates CORS when the PWA and Pro API share `https://axis.hoox.sh`. Cross-origin Pages → VPS still needs those headers (they are now default).

Venue sources/streams

Public Binance/OKX/etc. REST must send CORS headers usable by browsers. Firefox often reports “**CORS request did not succeed / status (null)**” when the TCP/TLS connection fails entirely (geo block, firewall, extension) — that is **not** a missing `Access-Control-Allow-Origin` header.

AXIS resilience for Binance:

1. Direct hosts: `api.binance.com` then `data-api.binance.vision` (`src/data/binance-http.ts`).
2. Worker allowlisted proxy: `GET /api/market/binance/klines|ticker/24hr|exchangeInfo` on the AXIS Worker (`worker/src/market.ts`).
3. Live WS rotates `:9443` → `:443` → `data-stream.binance.vision` (port 9443 is frequently firewalled).
4. Offline last resort: `mock-walk` / `mock-poll`, or warm bars-cache.

Prefer the DO relay (`cf-do stream / /api/stream`) when many tabs share one upstream WS — still requires the Worker edge to reach Binance.

AXIS resilience for MEXC:

1. Worker allowlisted proxy first: `GET /api/market/mexc/{klines,ticker/24hr,exchangeInfo}` (`src/data/mexc-http.ts`). Public `api.mexc.com` does **not** send `Access-Control-Allow-Origin`.
2. Direct `api.mexc.com` only as last-ditch fallback (offline lab, tests, or Worker 5xx). Worker 4xx is not retried against the CORS-blocked host.



Dynamic plugins

- **Module import** of plugin JS: needs CORS on the **script origin**. Prefer same-origin `/plugins/*` (PYNE Agent ships as `/plugins/axis-pine-agent.js`).
- **Remote** modules (e.g. legacy `pyne-agent-worker.../plugin/axis-pine-agent.js`) need ACAO + CSP `script-src` allow when a CSP is present.
- **fetch inside plugin**: subject to third-party CORS independently (agent API still uses the pyne-agent Worker origin).

Invariants & edge cases

1. **Credentials** — Worker CORS does not set `Allow-Credentials: true`; use Bearer headers, not cookies.
2. **Multiple prod domains** — product regex + comma-separated `ALLOWED_ORIGIN` cover multi-host; unknown sites never get open reflection.
3. **Vite vs 8081** — both covered for Worker; Flask config must match whichever you use.
4. **Health checks** from curl omit Origin — still work; browser calls need correct ACAO.
5. **No open `*.pages.dev`** — only the Pages project `axis` previews (`*.axis.pages.dev`) are product-scoped.

Failure modes

Browser console	Meaning
blocked by CORS policy	Origin not allowlisted
preflight 404	Server lacks OPTIONS
No ACAO on error body	Some error paths forgot headers (Worker try/catch mostly covered)

See also

- [Worker bindings](#)
- [Local dev](#)
- [Dynamic loader](#)

CI AND TESTING

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title: "CI and testing" description: "GitHub Actions for AXIS: unit coverage gate, Playwright smoke, worker typecheck, nightly full e2e."

CI AND TESTING

Abstract

AXIS quality gates live primarily under the repo root (Bun tests) and `worker/tests/`, orchestrated by GitHub Actions (`.github/workflows/ci.yml`, `axis-nightly.yml`). Deep test conventions are documented in [Testing reference](#) and `TESTING.md`.



Conceptual model

```
flowchart TB
    PR[Pull request] --> UNIT[Bun unit + worker tests]
    PR --> GATE[Coverage gate 95%]
    PR --> E2E[Playwright @smoke]
    PR --> WTC[Worker tsc]
    NIGHT[Nightly cron] --> FULL[Playwright all]
    NIGHT --> SEC[Security + coverage]
```

Interface surface — local commands

```
bun run test          # tests/ + worker/tests/
bun run test:unit
bun run test:coverage:gate # lcov + scripts/check-coverage.mjs 95
bun run test:security
bun run test:e2e:smoke
bun run test:e2e:critical
bun run test:e2e
bun run test:all      # gate + security
```

Worker:

```
cd worker
npm run typecheck
```

Repo Makefile:

```
bun run test:unit
make worker-typecheck
```

CI workflows

ci.yml (excerpted jobs)

Job (names vary)	What
Frontend tests / coverage	Bun install, coverage, artifact <code>axis-coverage-lcov</code>
<code>axis-e2e</code>	Playwright <code>@smoke</code> on PR
Worker typecheck	<code>worker</code> tsc
Related	Python suite remains for pynescrypt core



axis-nightly.yml

Job	What
Playwright full	all e2e specs
Security + coverage gate	test:coverage:gate + test:security

Schedule: `0 4 * * * UTC + workflow_dispatch`.

Internals

Path	Role
TESTING.md	Authoritative testing guide
scripts/check-coverage.mjs	Scoped line gate
playwright.config.ts	webServer build+preview :4173
e2e/	Specs with @smoke / @critical tags
worker/tests/	auth, keys, scripts, runtime
.github/workflows/ci.yml	PR gates
.github/workflows/axis-nightly.yml	Nightly

Coverage policy (summary)

- **Gate:** 95% lines on a **scoped** core (plugins, storage minus idb, store, results, sources, streams, data, selected chart helpers, worker auth/keys/runtime/scripts).
- **Excluded from gate:** heavy chart UI, pyodide boot, legacy JS, Solid `.tsx` surfaces (unit-tested elsewhere or deferred).

E2E design

- Builds production preview; baseUrl `http://127.0.0.1:4173`.
- Mocks `/run` and Binance where possible so smoke does not need live Pro API.
- Selectors use `data-testid` (`axis-btn-load`, `axis-manager`, ...).

Invariants & edge cases

1. **No real network in unit tests** — mock `fetch` / WebSocket.
2. **Import `./setup` first** when touching store/plugins.
3. **CI Bun version** pinned in workflows (e.g. 1.3).
4. **Playwright browsers** installed with `--with-deps` on CI.



Failure modes

CI red	Likely
Coverage gate	New untested code in scoped paths
Smoke timeout	Build slow; webServer 180s
Worker tsc	Env typing / DO exports
Flaky e2e	Missing testid; race without mock

See also

- [Testing reference](#)
- [Local dev](#)